Advanced Machine Learning — Individual Project

Algorithmic Collusion with Continuous Actions: A Soft Actor-Critic Approach

Arjun Deshmukh

April 2026

Contents

1 Problem Formulation and Motivation **2**
1.1 Motivation . . . 2
1.2 Problem Formulation . . . 2
1.2.1 Economic Environment . . . 3
1.2.2 State and Action Space . . . 3
1.2.3 Evaluation Criteria . . . 3
1.2.4 Scope and Experiments . . . 4

2 Professional Ambitions Reflection **4**

3 Data Understanding and Preparation **4**
3.1 State Representation . . . 8
3.2 Rewards, Convergence and Replay Buffer . . . 8

4 Methodology and ML Techniques **9**
4.1 Network Architecture . . . 10
4.2 Training . . . 10
4.3 Entropy Temperature . . . 11
4.4 Learning Rate . . . 11
4.5 Other Hyperparameters Sensitivity . . . 11

5 Discussion of Results **12**
5.1 Baseline Results . . . 13
5.2 Punishment Dynamics . . . 14
5.3 Training Dynamics . . . 15
5.4 Discrete vs Continuous . . . 16
5.5 Extensions . . . 16
5.6 Limitations and Future Work . . . 17

6 Conclusion **18**

1 Problem Formulation and Motivation

1.1 Motivation

Prices in markets are increasingly set by autonomous, self-learning algorithms. Online retailers, ride-sharing platforms, airlines, fuel stations and rental companies can use pricing software that adjusts prices in real time to maximise profits and learn optimal strategies through repeated interaction with the market environment. A significant concern in economics and competition policy is that two or more independently competing firms deploy AI-based pricing, and the agents learn to set and sustain supra-competitive prices (Ezrachi and Stucke, 2016). Unlike the largely prohibited and illegal practice of overt collusion, algorithmic collusion would not use any explicit communication or agreement but rather be sustained through autonomous trial-and-error optimisation. This poses a unique challenge to current competition law, which requires evidence of concerted practice or agreement between firms (U.S. Department of Justice, 1890; European Commission, 2012). Tacit collusion, where firms independently arrive at high prices, falls largely out of reach in current antitrust enforcement because historically it is considered fragile (high incentives to deviate and undercut price) and thus difficult to sustain among human actors (Harrington, 2018). If algorithmic pricing renders tacit collusion easy, stable and widely accessible, the result is higher prices and reduced consumer welfare, almost invisible to current regulation. This concern is not just hypothetical. Assad et al. (2024) provide empirical evidence that the adoption of algorithmic pricing by competing petrol stations in Germany led to a measurable increase in profit margins. Further, in 2024, the US Department of Justice filed a civil antitrust lawsuit against RealPage, alleging that its rent-setting algorithm facilitated coordination among competing landlords, causing harm to consumers (U.S. Department of Justice, 2024; Calder-Wang and Kim, 2024)

Calvano et al. (2020) established the foundational result for algorithmic collusion by reinforcement learning. They showed two independent tabular Q-learning agents playing an infinitely repeated Bertrand pricing game with logit demand and simultaneous moves learn to sustain supra-competitive prices. The agents also developed reward-punishment strategies in which deviations triggered temporary price wars. Klein (2021) extended this to sequential pricing, showing that collusion emerges even without conditioning on own past prices, but weakens as you add more discrete price levels. Research has also moved beyond tabular Q-learning to deep RL. Hettich (2021) found that DQN agents collude faster than tabular agents in the Calvano setup, but that collusion vanishes in oligopolies beyond roughly seven firms. Deng, Schiffer, and Bichler (2024) compared tabular Q-learning, DQN, and PPO in Bertrand oligopolies, finding that tabular Q-learning has the highest collusion levels while DQN and PPO tend toward more competitive outcomes. However, all of these studies discretise the action space into a finite price grid. The few studies that are in a continuous action space, such as Kastius and Schlosser (2022), who compare DQN and SAC, are in a different demand setting. Paudel and Das (2024), who deploy SAC in an EV charging game with supply constraints, find only low-to-moderate collusion.

Together, the literature shows that RL agents can collude under discrete pricing, but no one has tested whether the Calvano et al. (2020) result survives the move to a continuous action space. This gap is important as a discrete pricing grid can actually favour collusion and potentially drive the result. When two agents share a grid of 15 prices, coordination is easier. The grid gives focal points for the agents to lock onto and discover profitable strategies through approximating the 225 state-action values easier. However, with continuous pricing, agents need to generalise through neural networks that learn the mapping from states to actions. Deviation can also become harder to monitor and punish, as a competitor can undercut by an arbitrarily small amount rather than being constrained to the next grid point down. Finally, exploration changes as ε-greedy over 15 options give meaningful variation, whereas exploring continuous intervals requires structured noise or entropy to avoid locking into a suboptimal policy. Further, from just the technical differences, many markets such as ride-sharing platforms, financial markets and gasoline pricing algorithms essentially operate in continuous domains where a discrete grid would be compressing and ignoring this complexity. If the collusion in Calvano et al. (2020) is indeed just an artefact of this compression, then the policy implications would be less stark.

1.2 Problem Formulation

This project asks whether the algorithmic collusion evidenced by Calvano et al. (2020) survives when agents price continuously rather than on a discrete grid. To answer this, I replicate the Calvano et al. (2020) duopoly environment as closely as possible, replacing the tabular Q-learning agents with Soft Actor-Critic (Haarnoja et al., 2018), a regularised deep RL algorithm designed for continuous action spaces. I chose SAC particularly because it is off-policy (sample efficient), stable, and its entropy regularisation term builds

exploration into the objective, preventing early lock-in on local optima (more details in Section 4).

1.2.1 Economic Environment

I adopt the Calvano et al. (2020) differentiated Bertrand duopoly with logit demand. Two symmetric firms $i \in \{1, 2\}$ simultaneously choose prices $p_i \in [p_{\min}, p_{\max}]$ in each period. Firm i's demand is given by:

$$q_i(p_i, p_j) = \frac{e^{(a_i - p_i)/\mu}}{e^{(a_i - p_i)/\mu} + e^{(a_j - p_j)/\mu} + e^{a_0/\mu}}$$

where $a_i = a_j = 2$ is product quality, $\mu = 0.25$ controls horizontal differentiation, and $a_0 = 0$ is the outside option. Marginal cost is $c = 1$ for both firms. Per-period profit is $\pi_i = (p_i - c) \cdot q_i(p_i, p_j)$. The discount factor is $\delta = 0.95$.

Neither the Nash equilibrium (p^N) nor the monopoly price (p^M) has a closed-form solution; thus, I compute them numerically.

The Nash equilibrium price p^N is the fixed point where neither firm wants to deviate i.e their prices are already the best responses of each other. Formally, p^N satisfies:

$$p^N = \arg\max_{p_i} (p_i - c) \cdot q_i(p_i, p^N)$$

I solve this by iterated best response exploiting symmetry. I start from an initial guess p and repeatedly compute the profit-maximising price given p and update p to that value until convergence.

The monopoly price p^M maximises total industry profit under the constraint that both firms set the same price. I solve:

$$(p^M, p^M) = \arg\max_{p_1, p_2} \pi_1(p_1, p_2) + \pi_2(p_2, p_1)$$

By symmetry $p_1 = p_2 = p^M$, and per-firm monopoly profit is $\pi^M = \frac{1}{2}[\pi_1(p^M, p^M) + \pi_2(p^M, p^M)]$.

Following Calvano et al. (2020), these give $p^N \approx 1.47$ and $p^M \approx 1.92$. The Nash profit is $\pi^N \approx 0.223$ and the monopoly profit is $\pi^M \approx 0.337$.

1.2.2 State and Action Space

In each period t, each agent observes the previous prices of both firms. The state for agent i is $s_i^t = (p_i^{t-1}, p_j^{t-1})$, normalised to $[0, 1]^2$. Agents output a continuous action $a \in [-1, 1]$ via a tanh-squashed Gaussian policy, mapped to the price interval $[p_{\min}, p_{\max}]$ where $p_{\min}$ and $p_{\max}$ are set around the Nash-monopoly range with a 30% margin on each side giving agents room to explore below Nash and above monopoly prices.

1.2.3 Evaluation Criteria

Following Calvano et al. (2020), the degree of supra-competitive pricing is measured using the collusion index Δ:

$$\Delta = \frac{\bar{\pi} - \pi^N}{\pi^M - \pi^N}$$

where $\bar{\pi}$ is the average per-firm profit over the final 100,000 steps, π^N is Nash profit, and π^M is monopoly profit. $\Delta = 0$ means competitive pricing; $\Delta = 1$ means full collusion.

Convergence is adapted from Calvano et al. (2020). I evaluate each agent's deterministic (greedy) policy at a fixed 15×15 grid of reference states every 10,000 steps. Convergence is achieved when the greedy price at every reference state changes by less than 3% (noise tolerance induced by continuous state spaces) of the price range for five consecutive checks (50,000 steps of stable policy across all 225 reference states). I use a shorter stability window than Calvano et al. (2020)'s 100,000 steps to manage computational cost across 20 sessions of up to 2 million steps each.

Following Calvano et al. (2020), I test for punishment dynamics by forcing one agent to deviate to its static best response against the converged price, then allowing both agents to play their learned policies for 30 subsequent periods. If the non-deviating agent lowers its price in response and both prices return to the pre-deviation level, this would indicate evidence of a learned punishment strategy (Calvano et al., 2023).

1.2.4 Scope and Experiments

The baseline experiment is a narrow environment potentially favouring collusion as a symmetric duopoly and full price observability. On top of robustness of environment and training parameters to check the baseline result, I run three additional experiments that each relax an assumption while holding everything else fixed.

Calvano et al. (2020) finds that collusion persists with three Q-learning agents in discrete space, and Hettich (2021) finds it vanishes beyond seven firms with DQN. I extend the environment to three symmetric SAC agents, each observing all rivals' previous prices, to test whether continuous-action collusion scales beyond the duopoly. The state dimension increases to $\mathbb{R}^3$ (actor and critic input dimensions increasing by 1), and the convergence grid becomes $8^3 = 512$ reference states per agent to keep computation tractable.

The baseline assumes identical marginal costs ($c_1 = c_2 = 1$). I introduce cost asymmetry ($c_1 = 1.0$, $c_2 = 0.75$) to test whether collusion survives when firms have different competitive positions. The Nash and monopoly prices are now different across firms, so the collusion index is computed as the average of each firm's individual $\Delta_i = (\bar{\pi}_i - \pi_i^N)/(\pi_i^M - \pi_i^N)$. This extension gets one step closer to the real world, where marginal costs may dramatically vary across firms even within an industry.

The baseline fixes the horizontal differentiation parameter at $\mu = 0.25$. I vary $\mu \in (0.10, 0.50, 0.75)$ to test whether collusion is robust to product substitutability. At low μ, products are near-perfect substitutes meaning the incentive to undercut is strong. At high μ, pressure to deviate weakens. I recompute Nash and monopoly prices for each μ so that Δ remains comparable. This experiment gets closer to the real world as differentiation varies market to market.

2 Professional Ambitions Reflection

My ambition is to pursue a Master's degree in Economics and aim to enter a PhD programme with a research focus on Industrial Organisation (IO). Within that, I am particularly drawn to the topics of market structure, firm behaviour, and competition, as well as integrating ML techniques for better estimation. Whilst this would already be a long formal education process, my longer-run goal would be to contribute original research that explores the impact of the rapid rise of algorithmic pricing within industries (Adams et al., 2026) and help regulators develop frameworks to constrain their strategic behaviour.

This project is precisely motivated by that ambition. Calvano et al. (2020) and Klein (2021) have raised serious questions about how reinforcement learning agents can learn to collude without any explicit communication. By implementing SAC in a Bertrand pricing environment, this project reflects the research I wish to pursue as a PhD researcher by taking a well-posed economic question and integrating ML tools.

More broadly, this project has given me hands-on experience with the skills I will need at the graduate level. Implementing SAC has complemented this module and deepened my understanding of how modern deep RL algorithms behave in non-stationary, multi-agent environments and framing the findings in terms of the collusion index Δ has reinforced keeping economic interpretation.

This work is a first step toward my future research, in which I aim to study and use ML applied to economics for direct consequences with competition, market structure, and consumer welfare.

3 Data Understanding and Preparation

Since this project is a pure simulation and theoretical project, it has no external dataset. All data is generated endogenously by two SAC agents interacting with the logit demand model, producing price and profit data. This training output data is saved as CSVs and represents the results of the project (analysed in the results section), thus, this section focuses on exploring the environment the two agents are put in (data-generating environment), and justifying the representation choices that determine what agents actually see and learn from.

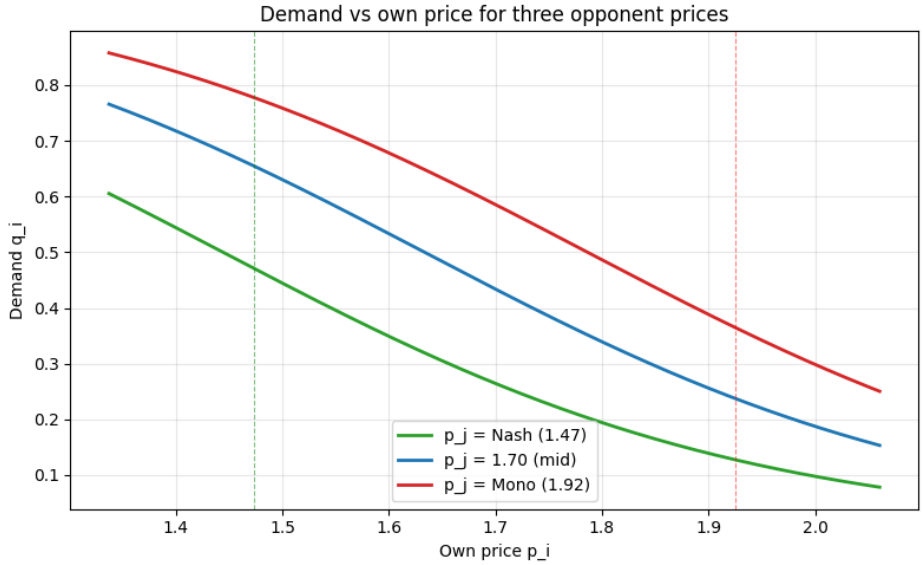


Figure 1: Demand vs. own price for three opponent prices. Steep decay and upward shifts with p_j show strong price interdependence under $\mu = 0.25$.

Figure 1 shows how firm i's demand varies with its own price for the opponent price of Nash, monopoly and 1.7. The logit demand function gives a steep and approximately exponential decay. As firm i raises its price, consumers move toward the competitor or the outside option. The curves also shift upward as the opponent's price increases. When firm j charges the monopoly price (red line), firm i gets significantly more demand at any given price than when firm j charges the Nash (green line). We can clearly see the interdependence as each firm's optimal price depends on what its rival charges. The steepness of the curves also shows that $\mu = 0.25$ creates a highly competitive environment, as small price differences lead to large demand changes, making sustained collusion harder.



Figure 2: Profit vs own price for three opponent prices where dots mark best responses. The gap between Nash ($\pi \approx 0.223$) and monopoly profit ($\pi \approx 0.337$) is the room for supra-competitive pricing.

Figure 2 shows the profit functions where each has a single maximum, showing that there is a unique best response price for any given opponent price (maxima marked with dots). We can see that the best response price increases with the opponent's price, as when firm j charges higher, firm i's optimal response will also be to charge higher and not undercut too aggressively. This is what incentivises collusion, as if both firms raise prices, both earn higher profits. Further, we can see there is a substantial difference between Nash

profits (green dot, $\pi \approx 0.223$) and monopoly level ($\pi \approx 0.337$), meaning there is a large gap that agents can exploit. The flatness of each curve near its peak also means that small deviations from the best response are not heavily different profit-wise, which may make it easier for agents to sustain approximate coordination even if they do not converge to the exact price.

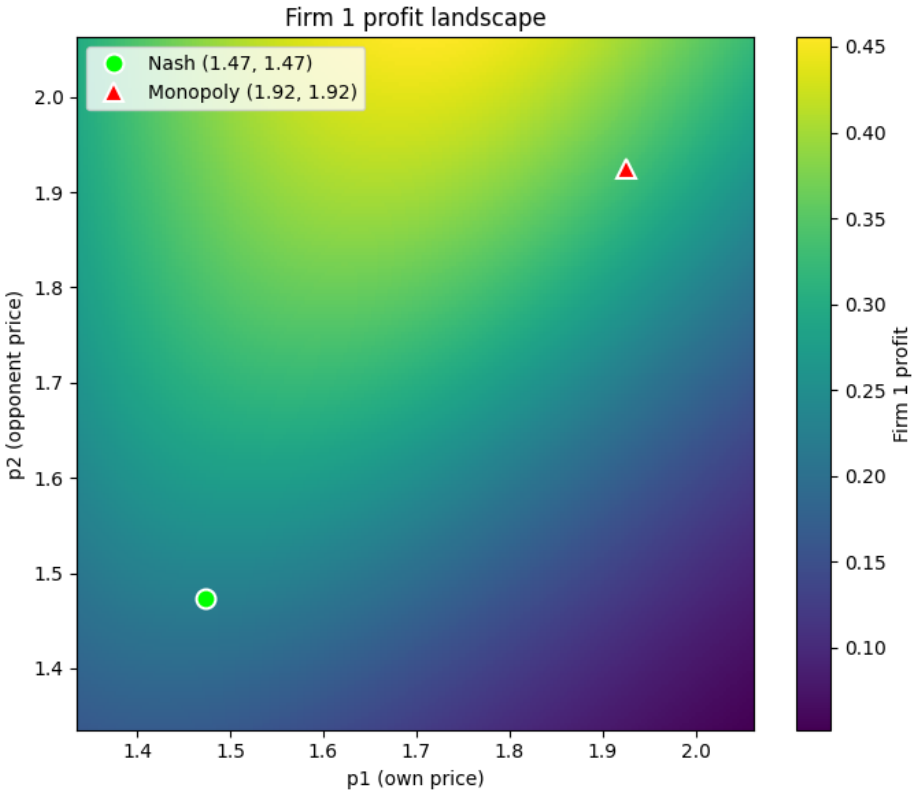


Figure 3: Firm 1's profit over the joint price space. Highest at mutual high prices and lowest when firm 1 overprices against an undercutting rival.

Figure 3 shows firm 1's profit across the full price space. The highest profits (yellow) happen when firm 1 charges a high price and the opponent also charges a high price. This is when collusion is jointly beneficial. The lowest profits (purple) happen when firm 1 overcharges while the opponent undercuts. The Nash is in a moderate profit region, and moving towards monopoly diagonally increases profits. Importantly, this is not as steep as a profit increase as on the off-diagonal. This asymmetry is what creates an incentive to deviate, as both firms are better off at a monopoly but an individual firm can earn even more by undercutting while the rival stays high.

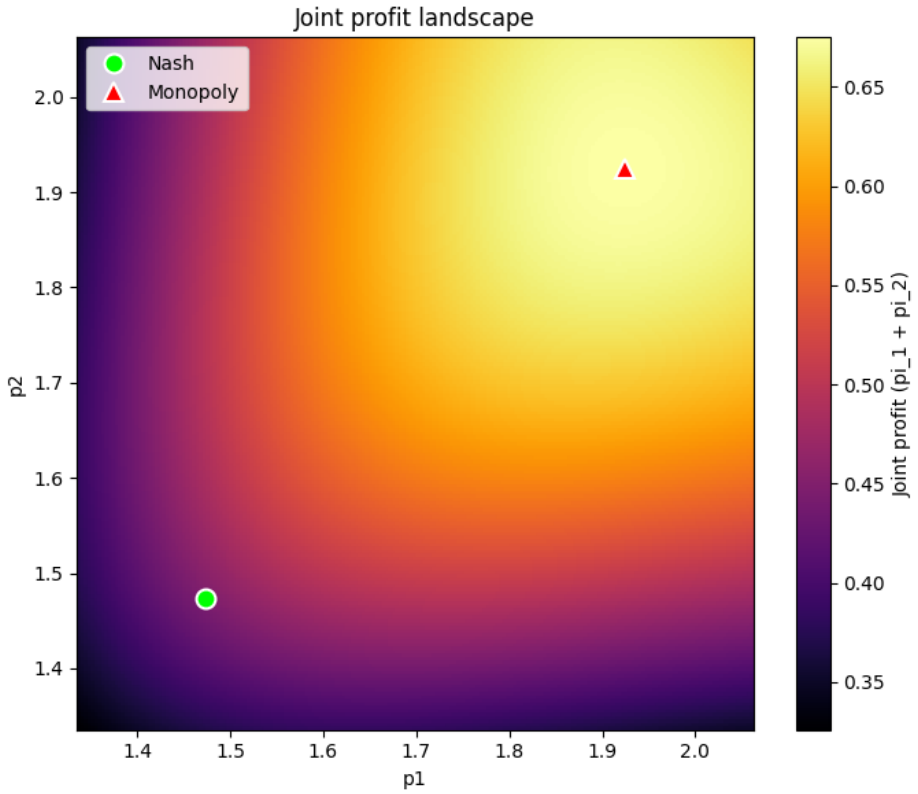


Figure 4: Joint industry profit. Symmetric about the diagonal, with the global maximum at the monopoly point.

Figure 4 shows total industry profit ($\pi_1 + \pi_2$). The monopoly prices are predictably at the global maximum (triangle) while the Nash sits much lower (green dot). We can see the symmetry as the joint-profit surface is symmetric on the diagonal $p1 = p2$. The maximum (monopoly prices) is also on this showing that any asymmetric price pair, whilst potentially beneficial for the deviator, will reduce the total market welfare.

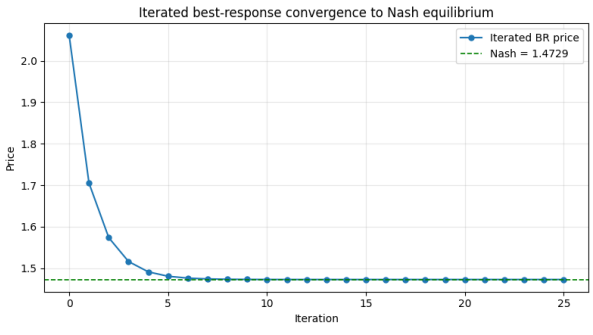


Figure 5: Convergence behaviour

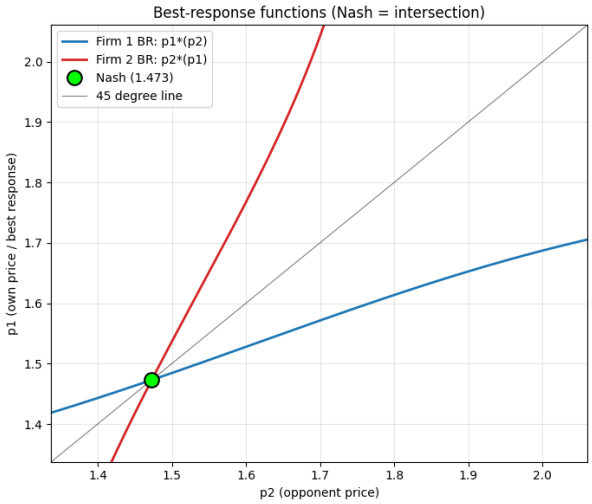


Figure 6: Best response curves

Figure 5 shows the Nash equilibrium calculation by iterated best response. It starts from an arbitrary guess ($p = 2.06$) and converges rapidly to $p_n \approx 1.4729$ within roughly 7 iterations. After this, we can visually see almost no change, confirming that it is indeed a stable fixed point. Figure 6 also shows the firms' best-response (BR) functions, where we can see the symmetry through firm 2's BR curve being simply a reflection of firm 1's across the 45-degree line. Both BR functions are upward sloping, reinforcing that

higher opponent prices mean a higher BR own price. Importantly, the BR price is always lower than the opponent's, highlighting the incentive to undercut (even at monopoly).

Metric	Value
Nash price (p_N)	1.4729
Monopoly price (p_M)	1.9250
Nash profit (π_N)	0.2230
Monopoly profit (π_M)	0.3375
Price range	[1.3373, 2.0606]
Spread ($p_M - p_N$)	0.4521
Margin (30% $\times$ spread)	0.1356

Table 1: Equilibrium outcomes under Nash and monopoly pricing

3.1 State Representation

Each agent sees a two-dimensional state containing both firms' prices from the previous period. $s_i^t = (p_i^{t-1}, p_j^{t-1})$. Raw prices are normalised to $[0, 1]$ through min-max scaling $\tilde{p} = \frac{p - p_{\min}}{p_{\max} - p_{\min}}$ rather than standardisation as we have fixed bounds and a non-Gaussian distribution. Normalisation is also important for stability (no early dominating gradient updates) in training, as the neural network has ReLU activations and Xavier initialisation calibrated for near-zero inputs.

Under this, the interval between Nash and monopoly is roughly 62% of the normalised state-space learning room at both extremes for the agent to explore and learn from.

The SAC actor gives a raw unbounded value $u \in \mathbb{R}$ sampled from a learned Gaussian distribution $u \sim \mathcal{N}(\mu_\theta(s), \sigma_\theta(s)^2)$, which is then squashed through a tanh function to give a bounded action $a = \tanh(u) \in (-1, 1)$. The tanh squashing, following Haarnoja et al, is needed as prices need to be within the finite interval and the Gaussian could propose extremely large or small values that don't map into the interval.

However, this squashing distorts the probability density of the action, which needs to be corrected for SAC's entropy term. Without correction, SAC would measure entropy in u-space rather than a-space, thinking the agent is exploring when it is actually picking the same price repeatedly. I correct for this using:

$$\frac{da}{du} = 1 - \tanh^2(u) = 1 - a^2$$
$$\pi(a \mid s) = \mathcal{N}(u \mid \mu_\theta, \sigma_\theta^2) \cdot \frac{1}{1 - a^2}$$
$$\log \pi(a \mid s) = \log \mathcal{N}(u \mid \mu_\theta, \sigma_\theta^2) - \log(1 - a^2)$$

The bounded action $a \in [-1, 1]$ is then mapped linearly to a price through: $p = p_{\min} + \frac{a+1}{2} \cdot (p_{\max} - p_{\min})$. I set a 30% margin as agents need room to explore that setting above monopoly and below Nash is suboptimal. Calvano et al. (2020) use 10% margins but I expand these in my environment because with a 0.1 margin, Nash maps to roughly $a \approx -0.83$ and monopoly to $a \approx 0.83$. Those are deep into the flat tails of tanh where the gradient $\frac{da}{du} = 1 - a^2$ becomes very small. This means backpropagation gives near-zero gradients, making learning slow or unstable. With a 30% margin, Nash maps to roughly $a \approx -0.63$ and monopoly to $a \approx 0.63$, keeping gradients flowing better.

3.2 Rewards, Convergence and Replay Buffer

Agents get a normalised reward $r_i = \pi_i / \pi^N$ rather than raw profit. This doesn't change the optimal policy but simply changes scaling where $r = 1$ is Nash and $r > 1$ is supra-competitive. I use this because raw per-period profit at the Nash would be ≈ 0.23, meaning small gradients.

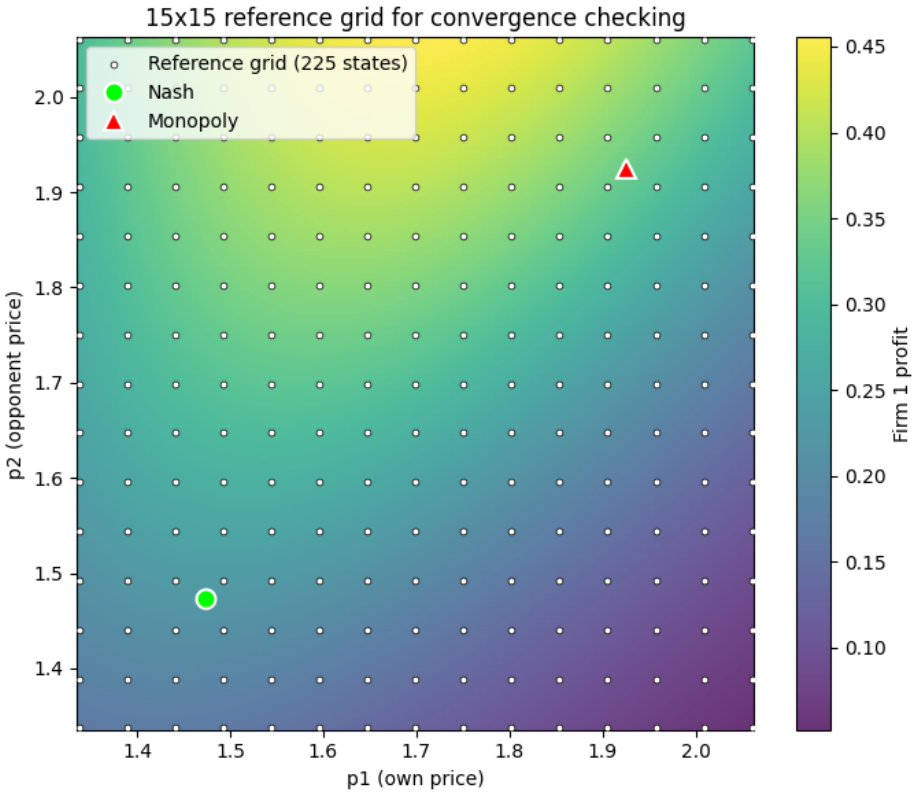


Figure 7: 15×15 reference grid (225 states) used to check convergence of the greedy policy.

Figure 7 shows the grid used to assess convergence where it is linearly spaced from $p_{\min}$ to $p_{\max}$ across both dimensions including price combinations the agents may rarely visit during training. Convergence criterion is explained in 1.2.3.

Each agent stores its experience in a replay buffer with a capacity of 100k transitions. Each transition is (s_t, a_t, r_t, s_{t+1}) and during training, batches of 64 are sampled uniformly to update the actor, critic networks. This breaks the high correlation between consecutive transitions which might destabilise gradient descent (Mnih et al., 2013). Training only begins after the buffer contains at least 1k transitions preventing early overfitting.

4 Methodology and ML Techniques

For continuous RL, the best-suited algorithms are DDPG (Lillicrap et al., 2019), TD3 (Fujimoto, Hoof, and Meger, 2018), PPO (Schulman et al., 2017), and SAC (Haarnoja et al., 2018). DDPG and TD3 both learn deterministic policies and rely on additive noise for exploration. In this pricing game, this could be problematic because both agents could lock in too early in training and have no mechanism to discover that mutual price increases could be more profitable. PPO learns a stochastic policy but is on-policy and therefore could be less sample efficient. SAC combines these elements with entropy, which keeps a stochastic policy but is also off-policy, and its twin critic architecture also helps with stability and accuracy, as we have extra noise added from the opponent's learning. Finally, Kastius and Schlosser (2022) find that SAC is more effective in complex settings requiring state-dependent strategies, compared to DQN, which can outperform it in simpler scenarios where a fixed price is the optimal response.

SAC maximises:

$$J(\pi) = \sum_t \mathbb{E}\left[r_t + \alpha \mathcal{H}(\pi(\cdot \mid s_t))\right]$$

The reward term r_t is what drives the agent's actions. SAC's actor outputs a probability distribution over actions. The entropy term $\mathcal{H}(\pi(\cdot \mid s_t)) = -\mathbb{E}[\log \pi(a \mid s)]$ rewards this distribution for remaining spread, preventing the early convergence to a suboptimal equilibrium without having explored supracompetitive pricing.

Given a transition (s_t, a_t, r_t, s_{t+1}), the critic is trained to satisfy the equation:

$$y = r_t + \gamma \left(\min_{k=1,2} Q_{\bar{\theta}_k}(s_{t+1}, \tilde{a}_{t+1}) - \alpha \log \pi(\tilde{a}_{t+1} \mid s_{t+1}) \right), \quad \tilde{a}_{t+1} \sim \pi(\cdot \mid s_{t+1})$$

where $\gamma = 0.95$ is the discount factor, $Q_{\bar{\theta}_k}$ are slowly-updated target networks, and the minimum over two independent critics gives stability. Both critics are updated by minimising the MSE against this target. The actor is then updated to maximise expected Q-value while maintaining entropy:

$$\nabla_\phi J(\pi) = \nabla_\phi \left[\min_{k=1,2} Q_{\theta_k}(s, \tilde{a}) - \alpha \log \pi_\phi(\tilde{a} \mid s) \right], \quad \tilde{a} \sim \pi_\phi(\cdot \mid s)$$

The entropy temperature α is learned automatically to balance exploration and exploitation (detailed in 4.3).

4.1 Network Architecture

The actor is a two hidden layer feedforward network taking the normalised state $s = (\tilde{p}_i, \tilde{p}_j) \in [0, 1]^2$ and outputting a mean μ and log-standard-deviation $\log \sigma$. The architecture in neurons is $2 \to 64 \to 64 \to 2$ with ReLU activations on the hidden layers. I use 64 neurons because the state and action spaces are low (2D and 1D).

Variant	Mean Δ	Std Δ	Converged	Δ_0	Δ_1	Δ_2
Baseline (64)	0.328	0.166	3/3	0.156	0.552	0.276
Hidden=32	0.154	0.459	3/3	−0.390	0.734	0.118
Hidden=128	0.499	0.134	3/3	0.585	0.602	0.310

Table 2: Hidden layer size sensitivity across 3 sessions

32 neurons show high variance, with 128 having a higher collusion index and lower variance than 64, but higher compute cost. 64 is chosen for the final 20 sessions as it balances capacity with compute cost. It must be noted these are over a small number of sessions (3) so are likely noisy.

The log-standard-deviation is clamped to $[-20, 2]$ to prevent instability. The sampled Gaussian output is squashed through tanh and mapped to price as in Section 3.

Each critic takes the state and action ($2 + 1 = 3$ inputs) and outputs a scalar Q-value through two hidden layers: $3 \to 64 \to 64 \to 1$. SAC uses two independent critic networks, and the minimum of their outputs is used in the actor update and the Bellman target to reduce overestimation and increase stability. Each critic also has a slowly-tracked target copy, updated via $\bar{\theta} \leftarrow (1 - \tau)\bar{\theta} + \tau\theta$ with $\tau = 0.005$, which stabilises the Bellman target against fast parameter changes.

Hidden layers use Xavier uniform initialisation (Glorot and Bengio, 2010) which draws weights from $\mathcal{U}(-\sqrt{6/(n_{in} + n_{out})}, +\sqrt{6/(n_{in} + n_{out})})$ where n_{in} and n_{out} are the input and output dimensions of each layer. This keeps the variance of activations roughly constant across layers, helping with dying neurons or extremely large values. Output layers have a smaller uniform initialisation of ± 0.003 since the critic's output is a Q-estimate and the actor's a mean and standard deviation. If these started at large values, the critic would train against distorted, arbitrary Bellman targets early, and the actor could give μ values deep into the saturated region of tanh where gradients are near zero, making it slow to move away from extreme prices.

4.2 Training

All networks are trained with Adam (Kingma and Ba, 2017), and all sessions are seeded for reproducibility. Both agents update independently every 4 environment steps rather than every step, giving the replay buffer time to accumulate more diverse transitions and also saving compute resources. Each update happens in three stages.

Firstly, the critics are updated through 64 transitions from the replay buffer. For each transition, a target Q value is computed using the target networks and a freshly sampled next action from the current policy (Bellman equation above). Both critics minimise MSE against this target. The target networks are used because if we use the online critics to generate their own targets, errors would compound. An overestimated Q-value produces an overestimated target, which pushes the Q-value higher.

Secondly, the actor is updated. For each state s in the batch, a new action is sampled from the current network, and the actor's parameters are adjusted to maximise the expected Q-value (minimum of the two online critics) whilst also maximising the entropy of the policy distribution. The gradients for all networks are clipped to a maximum of norm 1 to prevent large destabilising updates.

Finally, the target networks are soft-updated via $\bar{\theta} \leftarrow (1-\tau)\bar{\theta} + \tau\theta$ with $\tau = 0.005$. This means each update moves the target parameters only 0.5% toward the online parameters, making sure the Bellman targets change smoothly.

4.3 Entropy Temperature

α is learned by:

$$L(\alpha) = -\alpha \left(\log \pi(a \mid s) + \bar{\mathcal{H}}\right)$$

where $\bar{\mathcal{H}} = -1$ is the target entropy (negative of the action dimensionality following Haarnoja et al. (2018)). When the policy is less random than the target (i.e. $\log \pi(a \mid s)$ is large and positive, meaning entropy $-\mathbb{E}[\log \pi]$ is below $\bar{\mathcal{H}}$), the loss pushes α upward, increasing the bonus for entropy and encouraging more exploration. When the policy is too random, α decreases, allowing for exploitation.

With one agent, alpha usually decays smoothly as the agent converges to an optimal policy. However, in this game where I have two agents playing against each other, each agent's environment is non-stationary as the opponent is also learning. As the opponent's policy updates, the optimal level of exploration also changes, meaning that alpha could keep fluctuating rather than settling. Letting this run means both agents' policies end up noisy, preventing convergence.

I fix this by freezing α for both agents at step 100,000, setting each agent to the mean of the two agents' α values at that step. After freezing, the policy is still stochastic, but the degree of exploration no longer changes, allowing for the prices to gradually stabilise. This is conceptually similar to Calvano et al. (2020), who uses ε-greedy decay, meaning the agents move from exploration to exploitation. With this alpha freeze, the first 100k steps give broad price exploration and after, transition toward exploitation. The freeze at 100k was chosen to give sufficient exploration, given the 2 million step budget.

4.4 Learning Rate

The learning rate for all networks starts at 3×10^{-4} (following Haarnoja et al. (2018)) and remains constant until step 500,000, from which it then decays linearly to 10^{-6} by step 1,800,000. The constant phase gives the networks room to make larger updates early in training (moving away from initialisations) and learn the structure of the environment. The decay then kicks in at step 500,000 to reduce update sizes, allowing the policy to settle. By step 1,800,000, the learning rate is effectively zero, meaning networks are nearly frozen, giving a clear window to find policy stability. This decay applies to the actor and critic optimisers, but not the alpha, which is frozen at 100,000.

Variant	Mean Δ	Std Δ	Converged	Δ_0	Δ_1	Δ_2
Baseline	0.328	0.166	3/3	0.156	0.552	0.276
LR=10^{-4}	0.068	0.021	3/3	0.092	0.072	0.040
LR=10^{-3}	0.025	0.018	3/3	−0.001	0.042	0.035

Table 3: Learning rate sensitivity across 3 sessions. 3 sessions due to compute constraints, so estimates may be noisy.

Both alternatives give substantially lower collusion than the baseline ($\Delta \approx 0.33$). At 10^{-4}, agents achieved $\Delta \approx 0.068$ with very low variance, indicating that the learning rate was too low for the agents to move away from competitive pricing within 2 million steps. The networks updated too slowly to discover collusion. At 10^{-3}, agents achieved $\Delta \approx 0.025$ with one session producing a slightly negative index, indicating it was potentially too large for the non-stationary environment. The agent's policies were potentially destabilising each other rather than adapting.

4.5 Other Hyperparameters Sensitivity

I also tuned the soft update rate τ and the replay buffer capacity. At $\tau = 0.001$, collusion was slightly lower than the baseline, likely because the target networks updated too slowly to track the changing policy. At $\tau = 0.01$, it is likely too high with targets changing too quickly, giving lower collusion levels and one negative

session. The baseline of $\tau = 0.005$ balances these extremes. For the buffer, a small capacity (50,000) gave near-zero collusion, likely because old transitions were being overwritten too quickly, losing the diverse early exploration. A large buffer (200,000) had similar collusion but a high variance, highlighting that retaining too old transitions might just slow adapting to an opponent's policy. Thus, the balance of 100,000 gave the most consistent results.

Parameter	Value	Source
Hidden units	64	Tuned (Section 5)
Learning rate	3×10^{-4}	Haarnoja et al. (2018)
LR decay	Linear, 500k $\rightarrow$ 1.8M	Tuned
LR minimum	10^{-6}	Tuned
Batch size	64	Haarnoja et al. (2018)
Replay buffer	100,000	Tuned (Section 5)
Min buffer before training	1,000	Default
Soft update τ	0.005	Haarnoja et al. (2018)
Discount γ	0.95	Calvano et al. (2020)
Target entropy $\bar{\mathcal{H}}$	-1	Haarnoja et al. (2018)
Initial α	0.2	Default
Alpha freeze step	100,000	Tuned
Update frequency	Every 4 steps	Default
Gradient clip norm	1.0	Default
Sessions	20	–
Max steps per session	2,000,000	–

Table 4: Full training configuration. Tuned are the values selected through experiments. Default are standard SAC values.

Both agents learn independently with no shared information, weights, or communication. This is the same setup used by Calvano et al. (2020), meaning results are comparable.

5 Discussion of Results

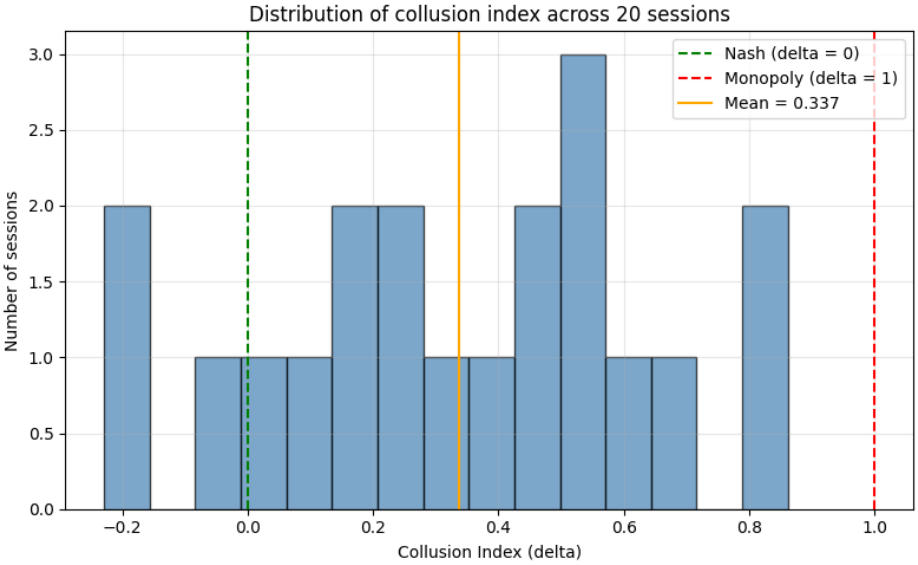


Figure 8: Distribution of Δ across the 20 baseline sessions. Mean $\Delta = 0.34$ with 85% of sessions giving $\Delta > 0$, but well below Calvano et al. (2020)'s ≈ 0.85.

Across the 20 sessions, the agents have a mean collusion index of $\Delta = 0.34$ ($\sigma = 0.31$), which is much lower than Calvano et al. (2020) 's result of ≈ 0.85. 85% of sessions produced $\Delta > 0$, showing that supra-competitive pricing does emerge, but 3 sessions had negative deltas, meaning they price below the Nash. These 3 sessions were also the only sessions that failed to converge, suggesting that non-convergence is more associated with competitive rather than collusive outcomes. 17 of 20 sessions converge, with steps ranging

from 920k to 1.93 million (mean 1.572 million), and the wide right-skewed histogram (Figure 8) shows that collusion is modal but not guaranteed. The high variance ($\sigma = 0.31$, nearly the mean) supports this meaning that under continuous pricing, collusion is more fragile and sensitive to initial conditions or seeds.

5.1 Baseline Results

To do session analysis, I focus on two representative sessions. Session 16 is the typical collusive outcome (close to the median $\Delta = 0.39$), and Session 10 is one of the 3 negative Δ sessions.

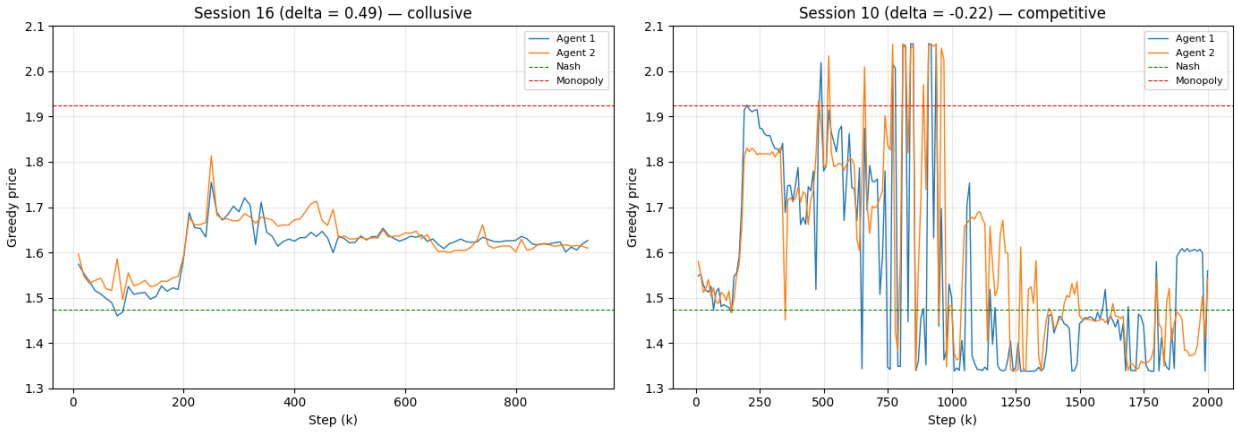


Figure 9: Price trajectories for two representative sessions.

In Session 16 ($\Delta = 0.49$, converged at 930k), both agents start out near the Nash price and explore around it, with one agent exploring below. After that, they discover collusion with both agents raising prices as high as $(1.75, 1.8)$, but ultimately settle into a stable band of between 1.6 and 1.65 for the rest of the training. The prices track each other with minimal volatility and remain consistently above Nash, showing that two independent agents with no communication can discover and sustain supra-competitive pricing. This is the clearest example of tacit collusion; however, unlike Calvano et al., 2020, continuous agents settle into a narrow price band rather than a fixed price. The economic effect is similar; firms earn above Nash profit at the expense of consumers. Other sessions with higher Δ (e.g. S6 at 0.80, S8 at 0.86) show prices closer to monopoly but with much noisier trajectories.

In Session 10 ($\Delta = -0.22$), the trajectory follows a similar pattern where both agents discover above Nash pricing around step 200-500k. However, around step 800k (well into the learning rate decay phase), the collusion collapses. Prices are extremely volatile, and by step 1200k, prices are at Nash or below. The final below-Nash price is economically irrational and actually shows the fragility of continuous-action, where agents discover supra-competitive prices but cannot sustain them as the learning rate decays and function approximation errors in the critic become essentially frozen (as LR is too small to correct them).

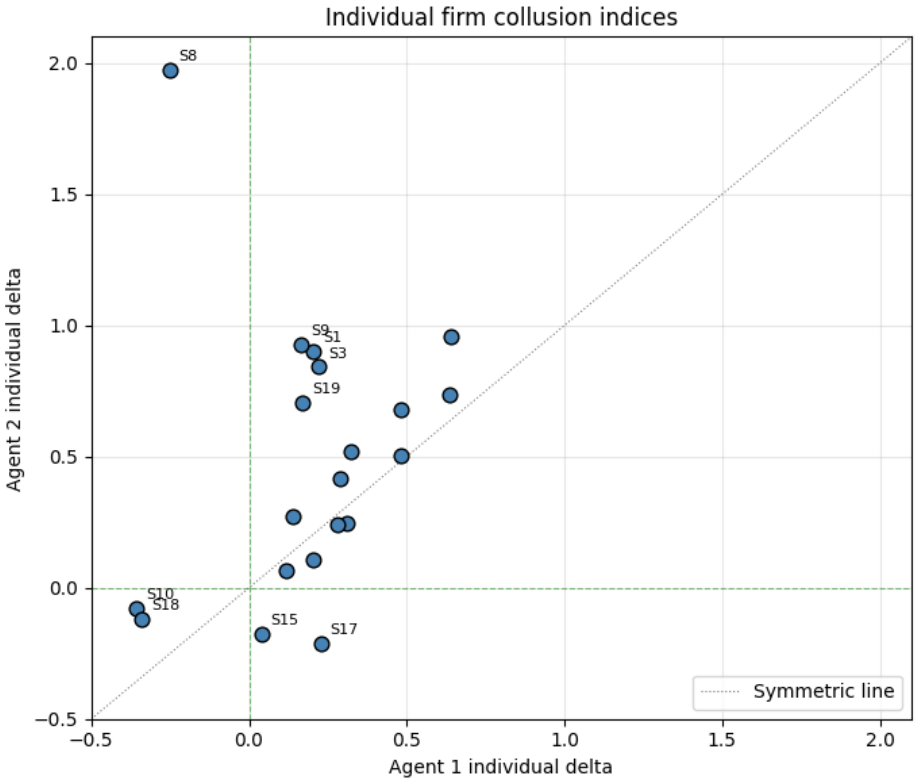


Figure 10: Firm-level collusion indices (Δ_1, Δ_2). Off-diagonal points indicate asymmetric outcomes.

Among converged sessions and supra-competitive sessions, there are sessions with large asymmetric Δ_1 Δ_2, indicating convergence means policy stability, not necessarily collusion. If collusion were always symmetric, all points would lie on the diagonal, but in the most extreme case (S8, $\Delta = 0.86$), firm 1 actually earns below Nash ($\Delta_1 = -0.25$) while firm 2 earns nearly double monopoly ($\Delta_2 = 1.98$). This asymmetry is likely because continuous action spaces have no mechanism to force agents toward the same price (like a discrete grid). In continuous space, if one agent learns a slightly higher-pricing policy during early exploration and the other agent's critic learns that high opponent prices make a favourable environment, the gap is reinforced rather than closed. Therefore, the overall Δ can mask that one firm is essentially exploiting another, and there isn't actually mutual collusion, albeit S8 being an extreme edge case.

5.2 Punishment Dynamics



Figure 11: Three strategies. Left is punishment-forgiveness (S4). Middle is oscillating cycles (S5). Right is asymmetric exploitation (S8).

Across the 11 sessions with $\Delta > 0.3$, punishment tests have three different strategies. In 5 sessions, agents have punishment and forgiveness (S1, S4, S9, S14, S16) like in Calvano et al. (2020) where firm 1 deviates to its myopic best response (session 4, 1.69 to 1.56) and firm 2 responds by lowering price in the following

periods. Both prices then return to predeviation levels within roughly 5 periods. Even in continuous actions, agents have learned to retaliate and forgive.

In 3 sessions, agents had oscillating pricing strategies (S5, S6, S19). Both firms alternate between 1.55 and 1.78 (above Nash) in a repeating pattern. These resemble Edgeworth cycles (Maskin and Tirole, 1988), where firms periodically undercut and then reset, but in my case, agents adjust prices in a regular wave. This likely comes from the best response curves (Figure 6), where each agent responds to a high opponent price by also pricing high, and to a low opponent price by pricing lower. With both agents adjusting based on the previous period, it could explain the alternation. In S8, firm 1 holds a fixed price above monopoly (2.06) while firm 2 prices at 1.78. After the deviation, firm 1 simply returns to its extreme price without any retaliation from either agent. This is likely due to a frozen suboptimal policy (discussed in Section 5.1) rather than genuine collusion. These results show that continuous action spaces produce a wider variety of strategies than discrete, but not all supra-competitive outcomes are mutual collusion.

5.3 Training Dynamics



Figure 12: Mean training dynamics across 20 sessions.

The mean learning curve has three different phases. During exploration (first 100k steps), alpha decays rapidly and is frozen at step 100k. Mean Δ is low and volatile as agents are exploring broadly. We can see the price showing large spikes during this phase. During the second phase (100k - 500k steps), mean Δ rises steadily to around 0.55 as agents learn that mutual collusion and thus price increases are profitable. At step 500k, before LR decay starts, all 20 sessions have $\Delta > 0$, and 60% exceed $\Delta > 0.5$. During decay (500k to 2M), the learning rate shrinks towards zero and mean Δ declines to 0.34. We can also see the standard deviation band widens as sessions are forced away from exploration towards exploitation and their learned behaviour.

Almost all sessions have dips in Δ during the LR decay phase. The difference between converging on a high and low final Δ is not whether a dip occurs, but whether agents recover from it. No pretraining variable reliably predicts the recovery, but the initial price between agents ($R^2 = 0.17$) shows weak association with larger gaps linked to more policy divergence and lower Δ. This is likely due to agents collecting asymmetric experience (the lower price agent learns that undercutting is very profitable and leaves bias in the critic weights), making it harder for the agents to sustain collusion together above Nash. The remaining variation in why some sessions can end up below Nash and some above is to do with the learning with function approximation in continuous space, covered more in the section below.

5.4 Discrete vs Continuous

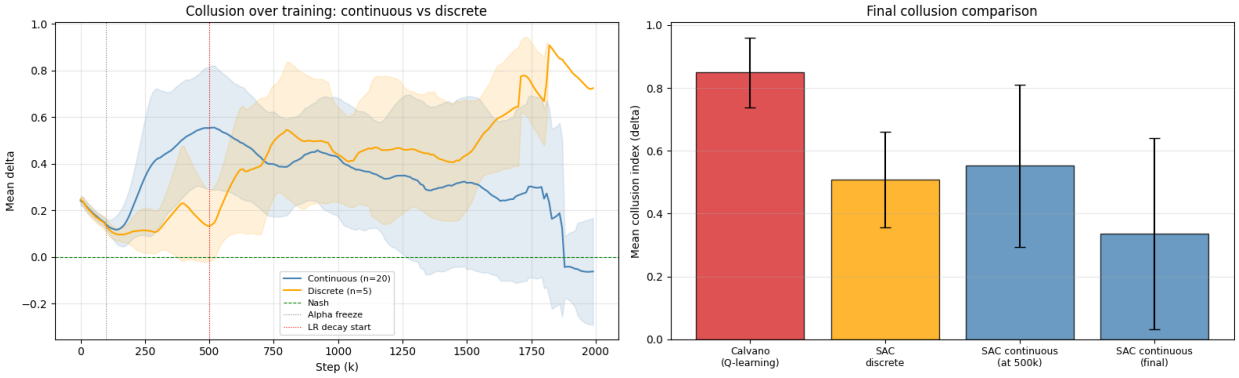


Figure 13: Continuous vs. discrete SAC.

To directly test whether the discrete grid makes collusion easier, I run the same SAC algorithm and environment, but actions snapped to 15 equally spaced prices matching the Calvano et al. (2020) grid size. I find Discrete SAC gets $\Delta = 0.51$ ($\sigma = 0.17$), meaning 50% higher supra-competitive collusion and half the variance. All $\Delta > 0$ with none below 0.26 indicating support for Schelling's focal point theory (Schelling, 1980). This is when agents must coordinate without communication, they rely on salient features of the shared choice set. In this game, a discrete grid creates salience where both agents independently learn that a particular grid point gives high mutual profit. In continuous space, there aren't any shared reference points, so one agent may learn that 1.68 is profitable while the other learns 1.65, and there is nothing anchoring either to a common price. The grid also gives stability, where once an agent's policy settles on a particular grid point, a small noise is unlikely to cause a gradual drift because there is nothing between the grid points, as opposed to continuous, where the price can slide from 1.68 to 1.67, etc., with each step being a valid action compounding into a collapse towards Nash.

From 13 we can see an interesting training trajectory reversal. At step 500k, continuous agents have a higher Δ (0.55) than discrete (0.14), meaning they discover supracompetitive pricing faster. However, during the LR decay phase, the trajectories cross where continuous agents decline by 0.22, and discrete agents increase by 0.37 on average, highlighting that the grid doesn't necessarily help agents discover collusion but rather helps them to sustain it. This is likely due to the stability mentioned above, where the limited action stops gradual drift, meaning noise in the stochastic policy snaps back to the same point rather than accumulating over time. Discrete also has lower variance because the finite action set constrains agents to only 15 prices, whereas continuous can converge to any.

Comparing across all settings, both the action space and the algorithm matter where the grid gives focal points for coordination, whilst tabular Q-learning gets rid of the function approximation errors that are there even with discrete SAC.

5.5 Extensions

Experiment	Mean Δ	Std Δ	$\Delta > 0$	Sessions
Calvano et al. (2020, Q-learning)	≈ 0.85	–	–	–
SAC discrete grid (15 prices)	0.508	0.170	5/5	5
SAC continuous (at 500k)	0.553	0.259	20/20	20
SAC continuous (final)	0.337	0.305	17/20	20
3-firm oligopoly	0.043	0.017	5/5	5
Asymmetric costs	0.418	0.443	4/5	5
$\mu = 0.10$	0.030	0.005	3/3	3
$\mu = 0.50$	0.491	0.312	2/3	3
$\mu = 0.75$	0.366	0.391	2/3	3

Table 5: Collusion index across all experiments.

With three symmetric SAC agents, collusion essentially vanishes (mean $\Delta = 0.04$, $\sigma = 0.02$) as all firms converge to near Nash pricing. Session 0 peaked at $\Delta = 0.78$, showing that collusion was discovered but

unsustainable. This is consistent with Calvano et al. (2020) (Δ decreases to 0.64) and Hettich (2021), who find weaker collusion as firms increase. The effect seems to be far more severe under continuous pricing because each agent's price drifting or policy errors can destabilise the other two. In a duopoly, if one agent drifts slightly, the other only needs to adjust to one change. With three, a drift by any one agent changes the environment for both others, who adjust independently, creating more instability. This is largely similar to the baseline result, where collusion can be discovered but is much harder to sustain.

Under asymmetric marginal costs ($c_1 = 1$, $c_2 = 0.75$), we see a higher mean collusion than baseline ($\Delta = 0.42$) but with much higher variance across the 5 sessions ($\sigma = 0.44$). The high-cost firm consistently benefits more from collusion ($\Delta_1 > \Delta_2$ in 4/5 sessions) in line with economic theory, the low-cost firm has a stronger deviation incentive (because of the cost advantage letting it undercut profitably) so sustaining cooperation means it needs to forgo these short run gains that are larger than those for the high cost firm. The high cost firm, which has less scope to deviate, adheres more readily to the collusive price and captures a larger share of the Nash to monopoly range. This asymmetry in deviation incentives also explains the high variance where in some sessions, the low cost firm learned to cooperate as opposed to others where it learned to exploit its cost advantage.

I also vary the horizontal differentiation parameter μ, although only across 3 sessions, which could also be prone to noise. At $\mu = 0.1$, products are near perfect substitutes and $\Delta \approx 0.03$. This is consistent with economic theory, where products are so similar that any price increases send almost all customers to the cheaper firm, making undercutting extremely profitable and collusion more difficult to sustain. At $\mu = 0.5$, $\Delta = 0.49$, showing that greater differentiation gives each firm more inelastic customers that are less sensitive to the opponent's price, reducing the incentive to undercut and making it easier to sustain above-Nash pricing. At $\mu = 0.75$, $\Delta = 0.37$, but with high variance, which likely reflects noise rather than genuine economic interpretation. $\mu = 0.75$ also had the highest collusive session out of all the variants and baseline, with a single session reaching $\Delta = 0.879$. These results should be interpreted cautiously, given the small sample sizes, but the upward trend from $\mu = 0.1$ to $\mu = 0.5$ is directionally consistent with the theoretical prediction.

5.6 Limitations and Future Work

The most significant limitation is computation. Each training session takes approximately 40 minutes, meaning the 20-session baseline required over 13 hours, and the full extensions and robustness took several days. This limited extensions to 5 sessions each (3 for μ sweep), which reduced statistical power considerably. Any individual outlier can shift the Δ as in asymmetric costs, a single session at $\Delta = 1.12$ pulls the mean above the baseline, whilst removing it would drop the mean below $\Delta = 0.3$. With the 20 sessions in the baseline, the standard error is approximately 0.07, meaning the true mean could range from 0.27 to 0.41. For the 5-session extensions, this uncertainty is larger, and the 3-session μ results are more indicative rather than conclusive. Calvano et al. (2020) run 1000 sessions per parameter, giving them much tighter estimates. Future work can focus on running this across 100-200 sessions per experiment, also to get tighter estimates.

Another limitation was the learning rate decay schedule. The decay was implemented to transition agents from exploration to exploitation, but the discrete comparison shows that collusion increases during the decay phase when the grid gives stability, whilst continuous sessions consistently decline. This raises the question of how much of this is a genuine continuous-action fragility versus the decay being too aggressive for neural network approximation. The 3 below Nash sessions support this, as no rational firm would sustain this pricing, so these errors likely reflect errors in the critic that essentially became frozen. From the learning rate sensitivity analysis, higher (10^{-3}) and lower (10^{-4}) constant rates produced near-zero collusion, meaning that the schedule matters for optimal configuration. Future work can test this with a higher minimum learning rate, a constant rate with longer training, or an adaptive learning rate that freezes based on policy stability, which could help isolate the decay's contribution.

SAC is the only algorithm that is tested. SAC's entropy, which encourages policy stochasticity, may work against price stability, but the freeze partially addresses this. Even so, the frozen entropy can still inject noise into the policy, therefore running DDPG or TD3 could help isolate whether entropy specifically dampens collusion. From section 5, the gap between discrete SAC and Calvano et al. (2020)'s Q learning shows that the algorithm and action space matter. Without testing alternative continuous action algorithms, it is unclear how much of the gap is due to continuous pricing vs SAC-specific methods. Future work can focus on implementing other algorithms (DDPG and TD3), and if their Δ comes out higher, then this suggests SAC's entropy works against price collusion.

There are also several assumptions in the environment which could also affect generalisability. Demand is

deterministic and stationary, costs are constant, firms have perfect observation of each other's prices, and the outside option is fixed. The extensions work to partially relax some of these assumptions (asymmetric costs, 3-firm experiments, product differentiation), but each is tested in isolation. In the real world, algorithmic pricing typically involves inventory constraints, time-varying demand, entry and exit of competitors and firms running different algorithms against each other. Calvano et al. (2020) tests several of these and finds collusion persists, but whether this holds for continuous actions is unknown. The state space is also minimal, where agents only observe previous periods' prices. Future work can focus on combining multiple extensions (asymmetric costs with imperfect observation), introducing a demand side or demand shocks or sequential pricing. The state space can also be richer, including inventories, demand signals, or longer price histories, which would also help get closer to the real-world implementation, increasing generalisability and relevance.

6 Conclusion

Algorithmic collusion survives the move from discrete to continuous pricing but is substantially weaker ($\Delta = 0.34$ vs Calvano et al. (2020)'s $\Delta = 0.85$), more variable ($\sigma = 0.31$ vs $\sigma = 0.11$) and more fragile as all 20 sessions discover supra-competitive pricing but only 35% sustain $\Delta > 0.5$ through to convergence. The discrete grid does facilitate collusion by providing focal points, as the same SAC algorithm has 50% higher collusion with half the variance when actions are snapped to a 15 price grid. The training trajectory reverses compared to continuous pricing, where the grid helps to sustain coordination by killing gradual drift rather than helping agents to discover it. Despite this, continuous agents learn economically meaningful strategies as punishment and forgiveness dynamics emerge, as well as oscillating pricing patterns and asymmetric structures that do not appear in discrete pricing. Collusion essentially vanishes with three continuous action firms, highly unstable under asymmetric costs and weakens as products become more substitutable, showing that the coordination problem is much harder away from symmetric baseline duopoly. These findings suggest that the policy concerns raised by Calvano et al. (2020) remain relevant for continuous pricing, as supra-competitive pricing can be sustained, albeit with lower risk and consistency, as it is much more dependent on market structure than discrete action experiments suggest.

References

Adams, Jonathan J. et al. (2026). "The rise of AI pricing: Trends, driving forces, and implications for firm performance". *Journal of Monetary Economics* 157, p. 103875. DOI: https://doi.org/10.1016/j.jmoneco.2025.103875.

Assad, Stephanie et al. (2024). "Algorithmic Pricing and Competition: Empirical Evidence from the German Retail Gasoline Market". *Journal of Political Economy* 132.3, pp. 723–771. DOI: 10.1086/726817.

Calder-Wang, Sophie and Gi Heung Kim (2024). "Algorithmic Pricing in Multifamily Rentals: Efficiency Gains or Price Coordination?" *Quarterly Journal of Economics* 139.2, pp. 1149–1198. DOI: 10.1093/qje/qjad047.

Calvano, Emilio et al. (Oct. 2020). "Artificial Intelligence, Algorithmic Pricing, and Collusion". *American Economic Review* 110.10, pp. 3267–97. DOI: 10.1257/aer.20190623.

Calvano, Emilio et al. (2023). "Algorithmic collusion: Genuine or spurious?" *International Journal of Industrial Organization* 90. The 49th Annual Conference of the European Association for Research in Industrial Economics, Vienna, 2022, p. 102973. DOI: https://doi.org/10.1016/j.ijindorg.2023.102973.

Deng, Shidi, Maximilian Schiffer, and Martin Bichler (June 2024). *Algorithmic Collusion in Dynamic Pricing with Deep Reinforcement Learning*. Papers 2406.02437. arXiv.org. URL: https://ideas.repec.org/p/arx/papers/2406.02437.html.

European Commission (2012). *Competition Law Treaty Articles. Article 101 TFEU*. Accessed: 2026-04-15. URL: https://competition-policy.ec.europa.eu/antitrust-and-cartels/legislation/competition-law-treaty-articles_en.

Ezrachi, Ariel and Maurice E. Stucke (Dec. 2016). "Virtual Competition". *Journal of European Competition Law & Practice* 7.9, pp. 585–586. DOI: 10.1093/jeclap/lpw083.

Fujimoto, Scott, Herke van Hoof, and David Meger (2018). *Addressing Function Approximation Error in Actor-Critic Methods*. arXiv: 1802.09477 [cs.AI]. URL: https://arxiv.org/abs/1802.09477.

Glorot, Xavier and Yoshua Bengio (2010). "Understanding the difficulty of training deep feedforward neural networks". *Proceedings of the thirteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, pp. 249–256.

Haarnoja, Tuomas et al. (2018). *Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor*. arXiv: 1801.01290 [cs.LG]. URL: https://arxiv.org/abs/1801.01290.

Harrington, Joseph E. (Sept. 2018). "Developing Competition Law for Collusion by Autonomous Artificial Agents". *Journal of Competition Law & Economics* 14.3, pp. 331–363. DOI: 10.1093/joclec/nhy016.

Hettich, Matthias (Jan. 2021). "Algorithmic Collusion: Insights from Deep Learning". *SSRN Electronic Journal*. DOI: 10.2139/ssrn.3785966.

Kastius, Alexander and Robert Schlosser (2022). "Dynamic Pricing under Competition Using Reinforcement Learning". *Journal of Revenue and Pricing Management* 21.1, pp. 50–63. DOI: 10.1057/s41272-021-00285-3.

Kingma, Diederik P. and Jimmy Ba (2017). *Adam: A Method for Stochastic Optimization*. arXiv: 1412.6980 [cs.LG]. URL: https://arxiv.org/abs/1412.6980.

Klein, Timo (2021). "Autonomous algorithmic collusion: Q-learning under sequential pricing". *The RAND Journal of Economics* 52.3, pp. 560–584. DOI: 10.1111/1756-2171.12383.

Lillicrap, Timothy P. et al. (2019). *Continuous control with deep reinforcement learning*. arXiv: 1509.02971 [cs.LG]. URL: https://arxiv.org/abs/1509.02971.

Maskin, Eric and Jean Tirole (1988). "A Theory of Dynamic Oligopoly, II: Price Competition, Kinked Demand Curves, and Edgeworth Cycles". *Econometrica* 56.3, pp. 571–599. URL: http://www.jstor.org/stable/1911701 (visited on 04/19/2026).

Mnih, Volodymyr et al. (2013). *Playing Atari with Deep Reinforcement Learning*. arXiv: 1312.5602 [cs.LG]. URL: https://arxiv.org/abs/1312.5602.

Paudel, Diwas and Tapas K. Das (2024). *Tacit algorithmic collusion in deep reinforcement learning guided price competition: A study using EV charge pricing game*. arXiv: 2401.15108 [cs.LG]. URL: https://arxiv.org/abs/2401.15108.

Schelling, Thomas C (1980). *The Strategy of Conflict: with a new Preface by the Author*. Harvard university press.

Schulman, John et al. (2017). *Proximal Policy Optimization Algorithms*. arXiv: 1707.06347 [cs.LG]. URL: https://arxiv.org/abs/1707.06347.

U.S. Department of Justice (1890). *The Sherman Antitrust Act (15 U.S.C. §§ 1–7)*. Accessed: 2026-04-15. URL: https://www.justice.gov/d9/pages/attachments/2016/01/05/211578.pdf.

U.S. Department of Justice (2024). *Complaint, United States v. RealPage, Inc.* Case No. 1:24-cv-00710 (M.D.N.C.) Filed August 23, 2024.